Fitness Progress Prediction: A Machine Learning Approach

Surya Raj nath Atul Nikhil Rana
South Asian University

Abstract

This project applies machine learning to predict fitness progress, specifically caloric expenditure, using user biometrics. We compared Linear Regression and Random Forest models on the Gym Members Exercise Dataset, ultimately finding that a baseline Linear Regression model provided optimal accuracy (R^2 = 0.9803) due to the strongly linear nature of biometric relationships.

1. Motivation

The modern fitness industry generates massive amounts of biometric data through wearable technology and fitness tracking applications. However, many users struggle to interpret this data to forecast their actual physical progress. The motivation behind this project is to bridge the gap between raw health metrics and actionable foresight. By leveraging machine learning, we aim to predict fitness outcomes—specifically caloric expenditure—based on easily trackable variables such as session duration, resting heart rate, and workout type. This predictive capability can help users optimize their routines, set realistic goals, and maintain long-term motivation.

2. Literature Review

Recent advancements in wearable technology and physiological data collection have prompted significant academic research into using machine learning for personalized fitness tracking. A 2025 study published in the International Journal for Research in Applied Science and Engineering Technology (IJRASET) explored the efficacy of replacing traditional, generalized metabolic equations with predictive algorithms. The researchers evaluated models including Linear Regression, Random Forest, and XGBoost based on datasets containing parameters like heart rate, weight, and exercise duration. They concluded that machine learning overcomes the limitations by accurately accounting for individual physiological variability.

Similarly, a separate 2025 study detailing a "Calorie Burn Tracker Using Machine Learning" (IJARCCE) directly compared linear models against complex ensemble methods. The researchers found that because biometric inputs scale consistently with energy expenditure, Linear Regression provided optimal results with over 94

3. Data Description

a) Attributes and Visualization: The project utilizes the "Gym Members Exercise Dataset," comprising 973 complete records. Key attributes include continuous variables such as Session Duration (hours), Avg BPM, Resting BPM, Weight (kg), and the target variable, Calories Burned. Exploratory Data Analysis (EDA) was conducted to visualize variable relationships.

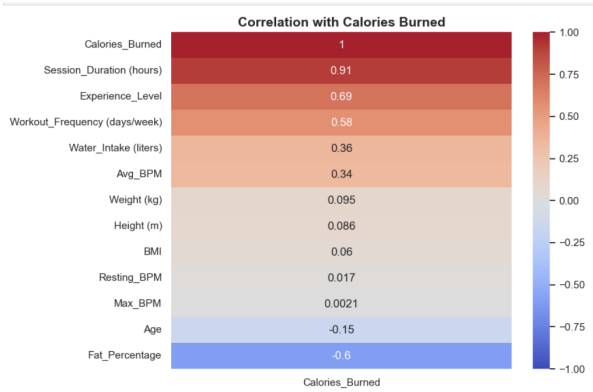

Figure 1. Correlation Heatmap of Fitness Variables

b) Preprocessing Details: The dataset contained zero missing or null values, ensuring high data integrity. Preprocessing involved isolating the target variable (Calories Burned) from the feature set. Categorical string attributes, specifically 'Gender' and 'Workout Type', were transformed into numerical formats using One-Hot Encoding. Finally, the data was partitioned using an 80/20 train-test split to allow for objective model evaluation.

4. Methodology

To establish a robust predictive pipeline, we implemented two distinct machine learning algorithms for comparison:

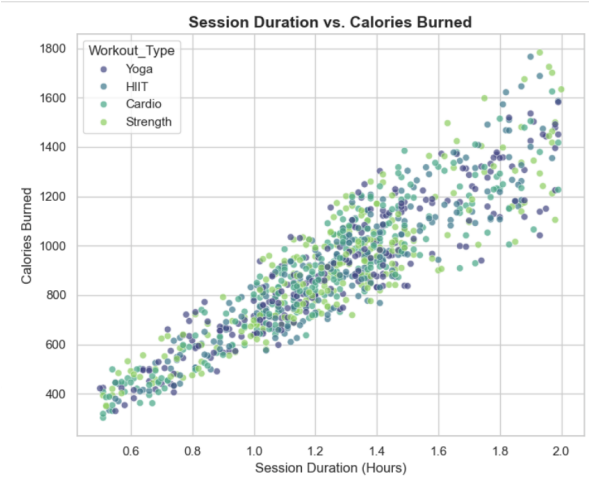

Figure 2. Session Duration vs Calories Burned

Baseline Model (Linear Regression): Selected due to the strong linear correlations observed during the EDA phase. This serves as our foundational metric for how standard biometrics scale linearly with calorie burn.

Advanced Model (Random Forest Regressor): An ensemble learning method utilizing 100 decision trees. This was implemented to capture any potential non-linear, complex relationships between variables that a simple linear model might miss.

5. Results and Analysis

To evaluate the predictive capabilities of our models, we utilized Mean Absolute Error (MAE) and the coefficient of determination (R^2). The baseline Linear Regression model achieved an R^2 score of 0.9803 and an MAE of 30.27. In contrast, the Random Forest Regressor achieved an R^2 of 0.9728 and an MAE of 36.49.

Interestingly, the baseline linear approach outperformed the ensemble method. This indicates that the features dictating our target variable share a strictly linear mathematical relationship. The Random Forest algorithm's tendency to partition data into non-linear decision trees resulted in slight overfitting, proving that for fundamental biometrics, simpler regression models are highly optimal.

6. Timeline

- **March 25 - March 28:** Dataset acquisition, environment setup, and Exploratory Data Analysis.
- **March 29 - March 31:** Data preprocessing, baseline model training, and advanced model training.
- **April 1 - April 5:** Interim report drafting and preliminary code pushed to GitHub.
- **Future Works (April - May):** Transitioning the model into a web application using Streamlit, allowing users to input live biometrics. We will also introduce a recommendation system to suggest workouts and supplements based on the predicted data.

7. Team Contribution

- **Surya (Lead ML Engineer):** Handled the core predictive pipeline. Responsibilities included writing the Python scripts for Exploratory Data Analysis (EDA), training both the baseline Linear Regression and advanced Random Forest models, and evaluating the final performance metrics.
- **Raj nath (Data Engineer):** Managed dataset acquisition and data preprocessing. Ensured the dataset was properly isolated, and applied One-Hot Encoding to categorical variables to prepare the data for algorithmic ingestion.
- **Atul (Research Lead):** Conducted the academic literature review. Synthesized recent peer-reviewed studies regarding machine learning in biometrics to validate the project's methodology and managed the strict CVPR document formatting.
- **Nikhil Rana (Product Architect):** Designed the theoretical framework for the project's next phase. Mapped out the architecture for the upcoming interactive web UI and formulated the logic for the rule-based recommendation system (diet and exercise suggestions).